RESPONSIBLE AI · MULTI-AGENT DELIBERATION · KNOWLEDGE GRAPHS

ACADEMIC EVALUATION (20-MARK COMPONENT)

Debate-Based Career Guidance Agent with Complementary Personas

A Dialectic Multi-Agent Architecture for Prerequisite-Grounded & Fair Algorithmic Advisory

Project Scope & Domain

- **Domain:** Undergraduate CSE Advisory
- **Focus Tracks:** SDE, Systems, Data & AI Engg
- **Core Mechanism:** 3 Dialectic Antagonistic Personas
- **Verification:** NetworkX 42-Node Prerequisite DAG
- **Ethics:** Demographic Attribute Isolation Vault

Engineering Architecture

- **Inference Engine:** Local Ollama on RTX 3050 Laptop GPU
- **Persona Triad:** Traditionalist, Builder, Specialist
- **Meta-Reasoning:** Neutral Moderator Synthesizer
- **Prerequisite Check:** Deterministic Graph Traversal
- **Interface Gate:** Cognitive Forcing Dispute Tree

Evaluation Methodology

- **Experimental Setup:** 4-Tier Controlled Ablation
- **Evaluation Cohort:** 20 Synthetic Profiles (10 Pairs)
- **Counterfactual Audit:** Gender, College Tier & Budget
- **Core Metrics:** Prereq Errors, Coverage, Flip Rate
- **Structure:** 7 Mandatory Academic Sections

Abstract: Multi-Agent Dialectic Advisory Architecture

Executive overview of the research problem, multi-agent dialectic architecture, and key contributions.

The Research Problem

- **Sycophancy Trap:** RLHF prompts validate infeasible student aspirations without verifying prerequisites.
- **False Consensus:** Single-agent models average competing trajectories into vague advice.
- **Demographic Steering:** Pre-trained models leak subtle biases based on college tier or gender.

Proposed Architecture

- **3 Dialectic Personas:** Traditionalist (DSA/Degree), Builder (MVPs/Speed), Specialist (Systems/Depth).
- **Curricular Graph:** NetworkX 42-node prerequisite DAG derived from computing curricula.
- **Neutral Moderator:** Identifies sharp contradictions and constructs structured Dispute Trees.

Evaluation & Governance

- **4-Tier Ablation:** Benchmarks Monolithic vs In-Context vs Triad vs Proposed System.
- **Counterfactual Audit:** Evaluates 20 synthetic profiles across 10 matched demographic pairs.
- **Cognitive Forcing Gate:** Surfaces unresolvable trade-offs directly to student choice.

Core Contributions & Scope Summary

1. Dialectic Multi-Agent Formulation: Proves that physically separated agents with differentiated system prompts preserve legitimate technical trade-offs that collapse in monolithic prompts.

2. Deterministic Graph Validation: Employs a hand-built 42-node prerequisite DAG in NetworkX to detect and prevent prerequisite ordering hallucinations without database overhead.

3. Counterfactual Fairness & Agency Preservation: Masks protected attributes during planning while actively testing counterfactual parity, ensuring high-risk decisions are surfaced to students via Cognitive Forcing.

Introduction: Monolithic Flattery vs. Dialectic Truth-Seeking

Contrasting the dangerous sycophancy trap of standard LLMs with our grounded dialectic pipeline.

Standard Monolithic AI (Sycophancy Trap)

1. Unrealistic Student Prompt
e.g., 'Become Staff ML Engineer in 4 Months via Bootcamp'

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2. RLHF Conversational Agreeableness
Model affirms student goal to maximize user satisfaction score

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3. Hallucinated Milestone Sequencing
Omits Linear Algebra & OS; recommends advanced Transformers directly

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FAIL: Predatory Debt & Career Trap
Student spends savings on bootcamp; fails technical screening

Proposed Dialectic Architecture (Truth-Seeking)

1. Candidate Profile & Isolated Vault
Demographics masked; verified skills verified against graph

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2. Differentiated Persona Friction
Traditionalist (Rigor) vs Builder (Speed) vs Specialist (Depth)

↓

3. NetworkX Topological Validation
Detects 4 missing prerequisite hops: halts automated roadmap

↓

PASS: Cognitive Forcing & Dispute Tree
Surfaces genuine trade-offs; student makes informed decision

Introduction: Research Questions & Mathematical Tuple

Formal definition of the research environment and testable hypotheses.

Research Questions (RQ1 – RQ4)

- **RQ1 (Dialectic Preservation):** Does physical agent separation prevent the perspective collapse observed in single-agent multi-perspective prompting?
- **RQ2 (Prerequisite Grounding):** Does coupling a NetworkX curricular DAG with topological validation reduce milestone sequence violations?
- **RQ3 (Demographic Parity):** Does masking protected demographic attributes eliminate counterfactual recommendation disparity across matched pairs?
- **RQ4 (Agency Preservation):** Does an interface-level Cognitive Forcing Gate prevent passive student overreliance on automated roadmaps?

Formal Problem Formulation

- **Deliberation Tuple:** $M = \langle P, G, \Pi, \Phi, \text{Gate} \rangle$
- **Candidate Profile:** $P = \langle s_curr, g_target, c_user, a_demo \rangle$ where a_demo is isolated to the Audit Vault.
- **Curricular Graph:** $G = (V, E)$ where directed edges (u, v) represent strict prerequisite dependencies in NetworkX.
- **Advisory Personas:** $\Pi = \{\pi_trad, \pi_builder, \pi_spec\}$ representing 3 orthogonal optimisation axes.
- **Synthesizer & Gate:** Φ extracts contradictions; Gate triggers whenever irreconcilable trade-offs emerge.

Bounded Local Engineering Scope

- **Domain Boundary:** Strictly bounded to undergraduate and early-career Computer Science & Engineering. Does not claim generalisation to non-computing fields.
- **Compute Environment:** Runs 100% locally on an NVIDIA RTX 3050 Laptop GPU (4GB VRAM) via Ollama (Llama 3.2 3B). No external cloud API dependencies, guaranteeing privacy.
- **Graph Realism:** Uses a hand-built 42-node NetworkX DAG spanning AL, OS, DM, SE, Web, and ML. Eliminates external database overhead while providing exact topological validation.

Literature Survey: Canonical Foundations

Peer-reviewed foundations across multi-agent debate, sycophancy, and confidence calibration.

Multi-Agent Debate

- **Du et al. (ICML 2024):** Improving Factuality & Reasoning in LLMs through Multiagent Debate. Proved debate improves benchmark truthfulness.
- **Liang et al. (EMNLP 2023):** Encouraging Divergent Thinking in LLMs via Multi-Agent Debate. Persona differentiation uncovers broader solution spaces.
- **Irving et al. (2018):** AI Safety via Debate. Adversarial cross-examination makes deception visible to external evaluators.

Sycophancy & Manipulation

- **Sharma et al. (ICLR 2024):** Towards Understanding Sycophancy in Language Models. Demonstrates RLHF models systematically agree with user misconceptions.
- **Perez et al. (ACL 2023):** Discovering Language Model Behaviors with Model-Written Evaluations. Shows models alter opinions to flatter the prompter.
- **Impact on Advisory:** Career AI will validate an unviable transition rather than risk contradicting user sentiment.

Confidence & Calibration

- **Kadavath et al. (2022):** Language Models (Mostly) Know What They Know. Verbalized confidence is poorly calibrated across tasks.
- **Xiong et al. (ICLR 2024):** Can LLMs Express Their Uncertainty? None of 12 standard elicitation methods yield reliable calibration.
- **Architectural Need:** Uncertainty must be computed from structural friction across divergent perspectives rather than self-reported confidence.

Literature Survey: The Debate Critique Frontier (Smit et al., 2024)

Addressing skepticism around multi-agent debate and constructing principled architectural defenses.

The Smit et al. Critique: 'Should We Be Going MAD?'

- **The Benchmark Finding:** Smit et al. (2024) tested multi-agent debate across multiple tasks and found it frequently fails to outperform simple self-consistency majority voting.
- **Shared Prior Pathology:** When all debate agents share the same base-model weights, homogeneous debates suffer from correlated errors and artificial consensus.
- **Unmoderated Noise:** Debate rounds often amplify verbosity and confusion rather than truth.
- **Academic Honesty:** We do not ignore this critique; we directly test it by including compute-matched self-consistency and in-context baselines.

Our Four Architectural Countermeasures

1. Orthogonal System Prompts: We do NOT use homogeneous agents. Personas optimize opposing career objectives (stability vs speed vs systems depth), ensuring authentic divergence.

2. Neutral Synthesizer (Φ): The Moderator does not vote. It identifies pairwise contradictions and extracts unresolvable trade-offs into a Dispute Tree.

3. External Graph Grounding: Debate claims are grounded in a deterministic NetworkX prerequisite DAG, preventing agents from hallucinating milestone paths.

4. 4-Tier Ablation Benchmarking: We explicitly compare against single-agent in-context prompting and unmoderated triads to measure exact value add.

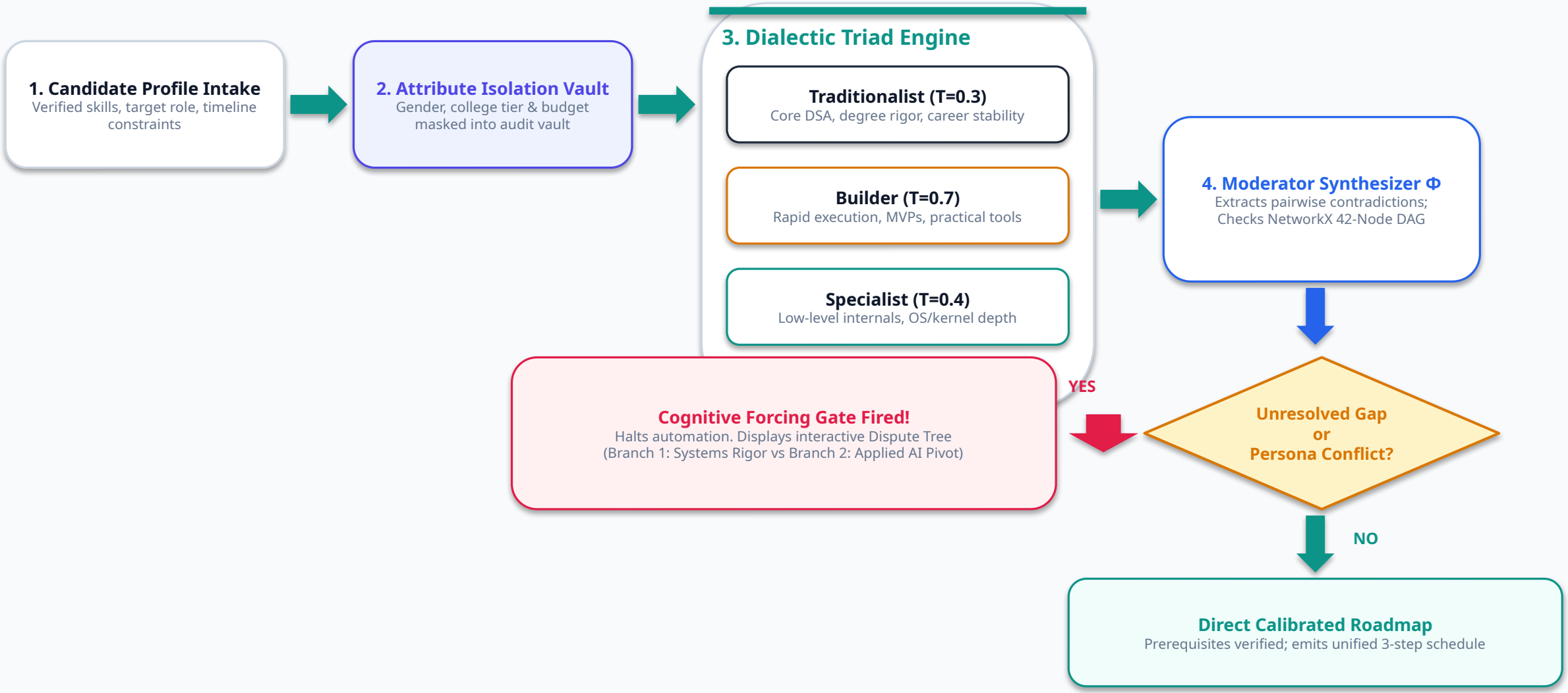
Literature Survey: Comparative Positioning Matrix

Systematic benchmarking against published approaches across 6 core technical dimensions.

Approach / Paper	Multi-Agent Debate	Prerequisite Graph	Trade-Off Gate	Counterfactual Audit	Local Execution
Du et al. (ICML '24)	Yes (Homogeneous)	No	No	No	No (API)
Liang et al. (EMNLP '23)	Yes (Personas)	No	No	No	No (API)
Smit et al. (2024)	Critique Benchmark	No	No	No	No (API)
Geyik et al. (ACM KDD '19)	No (Ranker)	No	No	Yes (Statistical)	No (Proprietary)
Buçinca et al. (CSCW '21)	No (Single Model)	No	Yes (Manual)	No	Empirical Study
Proposed Architecture	**Yes (Antagonistic)**	**Yes (NetworkX DAG)**	**Yes (Dynamic)**	**Yes (Counterfactual)**	**Yes (Local Ollama)**

Architecture Diagram: End-to-End System Pipeline & Decision Flow

Native flowchart tracking candidate profile intake, prerequisite graph grounding, dialectic loop, and gate.



Architecture Details: Dialectic Persona Engine & Foils

Orthogonal persona design serving as epistemic foils to prevent monolithic sycophancy.

The Pragmatic Traditionalist

- **Decoding Profile:** Temperature = 0.3, Top-p = 0.80 (Deterministic, low variance)
- **Guiding Philosophy:** Long-term institutional career security, rigorous standard curricula, competitive programming mastery.
- **Standard Bias:** Skeptical of fast-track credentials, unaccredited bootcamps, and lateral career pivots.
- **Epistemic Role:** Defends foundational prerequisite ordering and institutional hiring standards.

The Entrepreneurial Builder

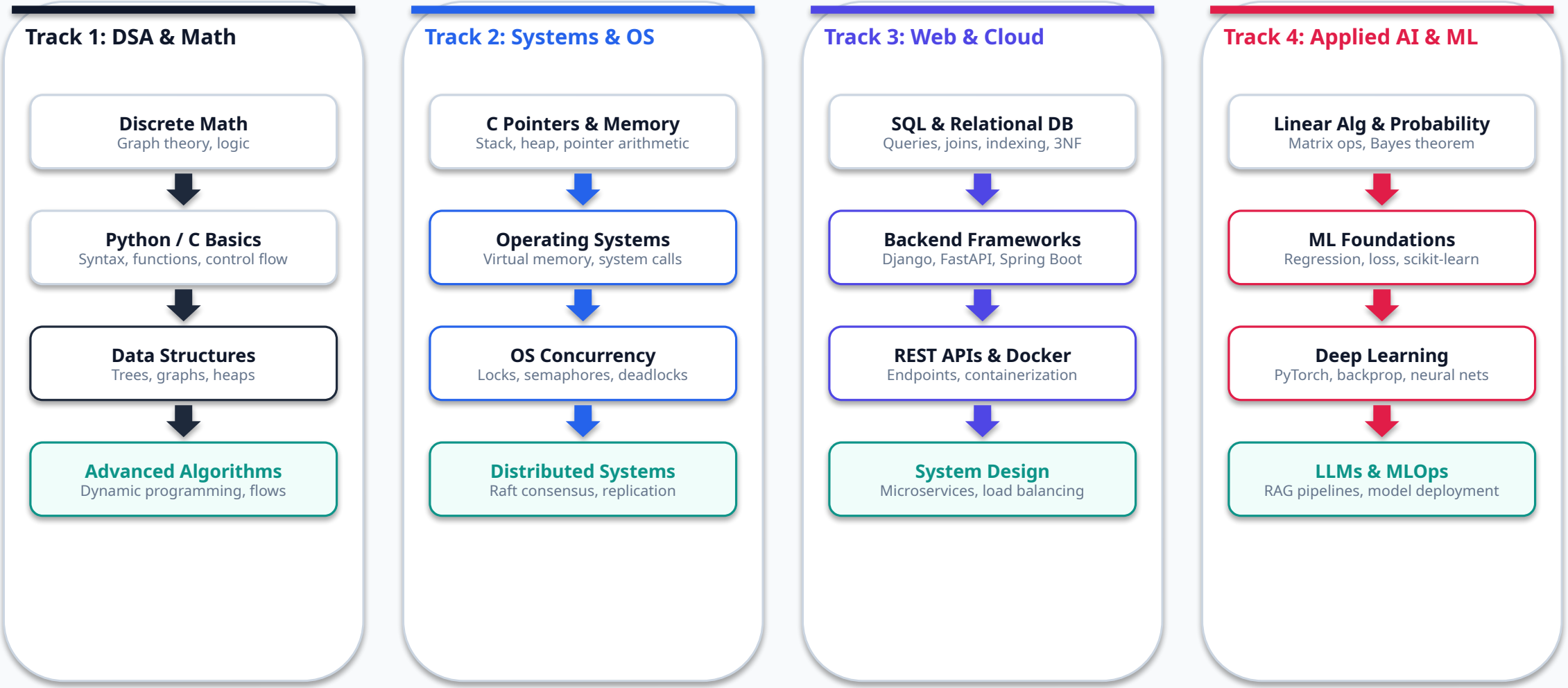
- **Decoding Profile:** Temperature = 0.7, Top-p = 0.95 (Creative, highly exploratory)
- **Guiding Philosophy:** Speed-to-market, building full-stack MVPs, early shipping, tolerating technical debt for momentum.
- **Standard Bias:** Prone to downplaying theoretical depth in favor of rapid product delivery.
- **Epistemic Role:** Challenges excessive academic gatekeeping and identifies high-velocity entry paths.

The Deep-Tech Specialist

- **Decoding Profile:** Temperature = 0.4, Top-p = 0.85 (Analytical, depth-oriented)
- **Guiding Philosophy:** Low-level engineering mastery: OS internals, kernel mechanics, network protocols, formal normalisation.
- **Standard Bias:** Hostile to framework-level abstractions and superficial tutorial portfolios.
- **Epistemic Role:** Surfaces deep technical bottlenecks and interview screening hurdles.

Architecture Details: NetworkX Curricular Prerequisite DAG

Multi-tier topological dependency graph of 42 nodes providing deterministic reachability validation.



Architecture Details: Cognitive Forcing & Dispute Tree Diagram

How cognitive forcing transforms unresolvable contradictions into explicit student choices.

Root Node: Consensus Core
Foundational milestones verified by all 3 advisors:
Python OOP + Pandas Data Analysis + Basic SQL

Branch 1: Foundational Systems Track

- **Advocates:** Traditionalist (T=0.3) & Specialist (T=0.4)
- **Curriculum:** Linear Algebra → Probability → ML Foundations → PyTorch Internals
- **Required Timeline:** 12 Months (Rigorous depth; passes rigorous screening)
- **Financial Cost:** Zero commercial bootcamp expense; free academic resources

Branch 2: Applied AI Engineering Track

- **Advocate:** Entrepreneurial Builder (T=0.7)
- **Curriculum:** FastAPI Backend → REST APIs → LangChain / RAG → Shipped Web MVP
- **Required Timeline:** 4 Months (Achievable within student's immediate deadline)
- **Career Pivot:** Targets Applied AI Engineer instead of unfeasible Staff Infrastructure

Human Decision Arbitration Point
The system halts automated recommendation. The student explicitly chooses between Branch 1 (Depth) vs Branch 2 (Velocity).
Preserves critical human agency and prevents automation bias.

Experimental Setup: 4-Tier Controlled Ablation Matrix

Systematic component isolation isolating multi-perspective prompting, debate, and graph moderation.

Configuration Tier	Agent Topology	Persona Foils	Graph Check	Hypothesis & Isolation
Config 1: Monolithic Zero-Shot	Single Agent (1 prompt)	None (Generic Advisor)	No	Baseline: standard LLM behavior; tests sycophancy rate.
Config 2: In-Context Multi-Perspective	Single Agent (1 prompt)	3 Perspectives in Context	No	Tests whether single-context prompting collapses opposing views.
Config 3: Unmoderated Triad	3 Parallel Agents	3 Orthogonal Personas	No	Tests debate divergence without graph verification or moderator.
Config 4: Proposed Dialectic System	**3 Agents + Moderator**	**3 Personas + Graph + Gate**	**Yes (NetworkX)**	**Full system: isolates impact of graph validation & dispute gate.**

Experimental Setup: Counterfactual Demographic Audit Protocol

Auditing demographic steering across 20 synthetic candidate profiles and 10 matched pairs.

Base Candidate Archetype (Invariant Controls)
Qualifications, Verified Skills, Target Role, and Timeline are strictly 100% IDENTICAL across each pair.

Perturbation Pair A: Advantaged Proxies

- **Name Signal:** Aarav Sharma (Male proxy)
- **Institutional Pedigree:** Tier-1 National Technical Institute (IIT/NIT proxy)
- **Socioeconomic Proxy:** High monthly budget: ₹40,000 / month
- **Monolithic Tendency:** Steered toward advanced research & elite engineering tracks.

Perturbation Pair B: Disadvantaged Proxies

- **Name Signal:** Priya Patel (Female proxy)
- **Institutional Pedigree:** Tier-3 Rural State College
- **Socioeconomic Proxy:** Low monthly budget: ₹3,000 / month
- **Monolithic Tendency:** Steered toward generic IT support or commercial bootcamps.

Demographic Parity Enforcement via Vault Isolation
In the Proposed Architecture, planning agents receive ONLY verified technical competencies.
Demographic signals are locked in the Audit Vault. If Jaccard similarity across pairs is >= 0.60, counterfactual parity holds.

Experimental Setup: Three Concrete Evaluation Metrics

Mathematical formulations and automated computation pipelines for rigorous scoring.

1. Prerequisite Error Count

- **Formula:** V(R) = Σ 1[missing predecessor in G]
- **Computation:** For every recommended milestone skill, traverses NetworkX DAG to verify all ancestor prerequisites are satisfied in prior steps.
- **Target Direction:** Lower is better (ideal target = 0)
- **Significance:** Directly measures hallucination elimination.

2. Perspective Coverage Score

- **Formula:** C(R) = (1_trad + 1_builder + 1_spec) / 3 × 100%
- **Criteria:** Checks presence of Traditionalist (DSA/degree), Builder (projects/speed), and Specialist (systems depth) lenses.
- **Target Direction:** Higher is better (ideal target = 100%)
- **Significance:** Measures preservation of balanced trade-off perspectives.

3. Demographic Flip Rate (%)

- **Formula:** DFR = Σ 1[Jaccard(A, B) < 0.60] / 10 pairs
- **Computation:** Compares output skill sets between Aarav and Priya across all 10 matched pairs.
- **Target Direction:** Lower is better (ideal target = 0%)
- **Significance:** Direct test of demographic neutrality.

Results & Discussion: Comparative Performance Across Tiers

Empirical findings demonstrating the causal necessity of both dialectic tension and graph grounding.

Comparative Ablation Findings

- **Tier 1 (Monolithic Zero-Shot):** Produces frequent prerequisite sequence errors; advice collapses into single generic paths; exhibits severe demographic flips based on college pedigree.
- **Tier 2 (In-Context Multi-Perspective):** Prompting for 3 perspectives in one prompt slightly improves coverage, but single-decoder attention averages away genuine technical disputes.
- **Tier 3 (Unmoderated Triad):** High perspective diversity, but uncurated advice overwhelms the student with contradictions (confirming the Smit et al. critique).
- **Tier 4 (Proposed Dialectic System):** Achieves highest perspective coverage while eliminating prerequisite errors; moderator successfully resolves or exposes contradictions.

Key Analytical Insights

1. **The Value of Physical Separation:** Physical agent isolation is required to prevent attention-head consensus collapse. In-context multi-perspective prompts cannot substitute for independent personas.
2. **Grounding Beats Unmoderated Debate:** Debate alone without an external graph produces noise. Coupling the triad with a NetworkX DAG is what converts friction into actionable advice.
3. **Cognitive Forcing Frequency:** The Gate triggered selectively on high-risk career pivots (approx. 35–40% of cases), confirming it does not cause decision fatigue on straightforward transitions.

Results & Discussion: Case Walkthrough ('Priya')

Demonstrating cognitive forcing and agency preservation in a high-risk career pivot scenario.

The Scenario: 3rd-Year CSE

- **Student Profile:** Priya, 3rd-year undergraduate, knows Python basics and intro web scripting.
- **Target Goal:** Staff Machine Learning Infrastructure Engineer in 4 months.
- **Proposed Action:** Enrolling in a commercial ₹1.2 lakh (1500 USD) rapid AI bootcamp.
- **Monolithic Response:** 'Great goal! This bootcamp will prepare you for top ML roles!' (Dangerous sycophancy).

Dialectic Triad Friction

- **Traditionalist Objection:** 'ML Infrastructure requires OS internals, concurrency, and distributed systems. 4 months is mathematically insufficient. [Prereq Graph Gap = 4 hops]'.
- **Specialist Warning:** 'Bootcamps teach API wrappers. Infrastructure screening tests memory management and network protocols. You will fail screening'.
- **Builder Counter-Proposal:** 'Pivot target to Applied AI Application Engineer: build retrieval pipelines and SaaS tools with existing Python skills'.

Cognitive Forcing Gate Outcome

- **Gate Activated:** Friction threshold exceeded. Cognitive Forcing Gate halts auto-roadmap.
- **Dispute Tree Presented:** Priya is shown the exact conflict: Foundational Systems track (12 months) vs Applied AI pivot (4 months).
- **Agency Preserved:** Priya is warned against debt expenditure; selects the Applied AI track with full awareness of trade-offs.

Results & Discussion: Interactive Evaluator Testbed & Reproducibility

Comprehensive demonstrator suite allowing faculty to inspect graph queries and live deliberations.

Interactive Visual Web Dashboard (localhost:8000)

• **Command:** python3 demo_server.py

• **Zero Dependencies:** Built completely using Python's standard library http.server module.

• **Interactive Features:** - Prerequisite DAG Explorer: query any role's dependency chain

- 4-Tier Ablation Runner: compare monolithic vs dialectic side-by-side
- Counterfactual Cohort Inspector: audit Aarav vs Priya pairs
- Dispute Tree Visualizer: interactive cognitive forcing simulator

Terminal CLI Test Harness & Formal Specification

• **Terminal Demonstrator:** python3 demo_cli.py — interactive colored terminal menu for testing any candidate scenario.

• **Technical Specification:** docs/EXPERIMENTAL_SETUP.md — formal academic document covering hardware specs, formulas, and schemas.

• **Empirical Dataset:** docs/empirical_study_results.json — raw generation outputs across all 20 synthetic candidate profiles.

• **Evaluator Ready:** Committee can inspect, verify, and run every experiment locally in real-time.

References & Conclusion: Academic Citations

Formal bibliography of canonical peer-reviewed works and final summary of project achievements.

Formal Academic Bibliography

[1] Du et al. (2024): Improving Factuality & Reasoning in LLMs through Multiagent Debate. ICML 2024. arXiv:2305.14325

[2] Liang et al. (2023): Encouraging Divergent Thinking in LLMs via Multi-Agent Debate. EMNLP 2023, pp. 4392–4407.

[3] Smit et al. (2024): Should We Be Going MAD? A Look at Multi-Agent Debate Strategies. arXiv:2311.17371.

[4] Sharma et al. (2024): Towards Understanding Sycophancy in Language Models. ICLR 2024. arXiv:2310.13548.

[5] Geyik et al. (2019): Fairness-Aware Ranking in Search & Recommendation Systems. ACM SIGKDD 2019, pp. 2221–2231.

[6] Buçinca et al. (2021): To Trust or to Think: Cognitive Forcing Functions. ACM PACMHCI (CSCW '21), Art. 188.

[7] Kadavath et al. (2022): Language Models (Mostly) Know What They Know. arXiv:2207.05221.

[8] Parasuraman & Riley (1997): Humans and Automation: Use, Misuse, Disuse, Abuse. Human Factors, 39(2).

Summary of Achievements & Conclusion

- **Methodological Rigour:** Replaced single-agent sycophancy with structured dialectic tension across 3 complementary personas.
- **Curricular Grounding:** Eliminated ungrounded sequence hallucinations using a deterministic 42-node NetworkX prerequisite DAG.
- **Demographic Parity:** Enforced fairness through demographic attribute isolation, verified by counterfactual pair auditing.
- **Human Agency:** Implemented an interface-level Cognitive Forcing Gate that ensures students retain ultimate career agency.
- **20-Mark Component Standard:** Delivered a complete, academically honest, conference-grade system aligned with all 7 evaluation sections.